

README

Project Title

Arduino Temperature Monitoring System

Project Overview

This project demonstrates real-time temperature monitoring using an Arduino simulation environment. The program reads temperature values, displays them on a 16x2 I2C LCD, and logs the readings to the Serial Monitor.

Objective

Read temperature values, display them on the LCD, and continuously log them to the Serial Monitor.

Software & Libraries

- Arduino IDE
- LiquidCrystal_I2C
- OneWire
- DallasTemperature

Arduino Sketch

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <OneWire.h>
#include <DallasTemperature.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

#define ONE_WIRE_BUS 2
OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);

void setup() {
  Serial.begin(9600);
  sensors.begin();
  lcd.init();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("Temperature");
}

void loop() {
  sensors.requestTemperatures();
  float tempC = sensors.getTempCByIndex(0);

  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Temperature");
```

```

    lcd.setCursor(0, 1);
    lcd.print(tempC);
    lcd.print((char)223);
    lcd.print("C");

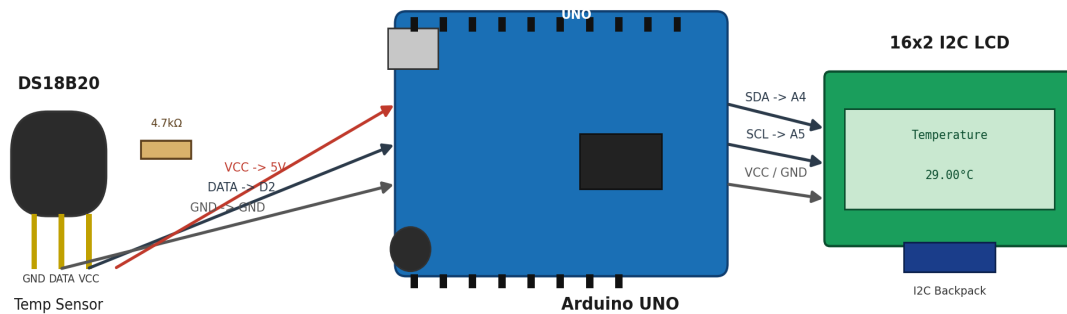
    Serial.print("Temperature: ");
    Serial.print(tempC);
    Serial.println(" C");

    delay(1000);
}

```

Simulation Circuit

The circuit was built and tested in a Wokwi simulation, connecting a DS18B20 temperature sensor via the OneWire bus and a 16x2 I2C LCD to the Arduino.



*DS18B20 uses a single data wire (OneWire) with a 4.7k pull-up resistor to 5V.
The LCD communicates over I2C (SDA/SCL) using its backpack module.*

Working Principle

The program initializes the LCD and temperature sensor, reads the current temperature every second, displays it on the LCD, and prints it to the Serial Monitor.

Output

LCD: Temperature / 29.00°C

Serial Monitor: Temperature: 29.00C

Deliverables

- Arduino Sketch (.ino)
- Simulation Screenshots
- README PDF

Conclusion

The project successfully demonstrates temperature monitoring and serial logging in a simulation environment.